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Capacity Allocation and Scheduling in Supply Chains

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We consider a multiple product supply chain where a manufacturer receives orders from several distributors. If the orders cannot all be met from available production capacity, then the manufacturer allocates that capacity and a set of resubmittable orders among the distributors. The distributors may share their allocated capacity among themselves before submitting revised orders. Finally, the manufacturer schedules the revised orders to minimize its cost. We consider three practical coordination issues. First, we estimate the benefit to the manufacturer from considering scheduling costs and constraints in making capacity and order allocation decisions. Second, we estimate the additional profit that the distributors achieve when they share their allocated capacity. Third, we estimate the value of coordination between the manufacturer and the distributors. Our work is among the first to consider all three issues simultaneously. We model scheduling costs and constraints within the manufacturer's capacity allocation problem. We model the distributors' capacity sharing problem as a cooperative game that has properties that are unique within cooperative game theory. Finally, we develop optimal algorithms for all the models defined by the three coordination issues. Our exact evaluation of decisions about the appropriate coordination level improves managers' ability to make those decisions.

Subject classifications: supply chain; capacity allocation, scheduling; cooperative game; value of coordination; production/scheduling: sequencing, deterministic, single machine; games/group decisions: cooperative.

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1. Introduction

In many industries, manufacturers respond to a purchase order that exceeds their scheduling capacity with an offer of partial fulfillment. This defines the following sequence of events. First, the distributors order more than the manufacturer can feasibly schedule. Second, based on its ability to schedule some of the orders, the manufacturer responds with an offer that specifies restrictions of the orders and/or of the available capacity. Third, the distributors revise their orders subject to these restrictions. The following three paragraphs document several specific applications of these three steps.

Capacity shortfall issues arise in many industries. Examples include industries with high fashion content, rapid technological development, or occasional demand surges (Fisher 1997). Iyer et al. (2003) identify several practical examples in the fashion goods, telecommunications, and electricity supply industries. Durango-Cohen and Yano (2006) point out that capital-intensive industries have to deal with both demand volatility and the inability to change their capacity quickly to match demand, which makes them vulnerable to capacity shortfalls.

When faced with orders that they cannot meet, manufacturers apply several capacity allocation mechanisms

in practice. One commonly used mechanism places distributors "on allocation." This means that specific items from the distributor's order are restricted by the manufacturer. Examples exist in numerous industries, such as soft drinks (Kherrmouch 1998), semiconductors (*EE Times* 2006), and steel (*New South Construction Supply eNews* 2008). A second mechanism that achieves the same effect is the use of price changes to adjust demand for specific items. This mechanism is applied to the supply of popular toys, DRAM chips, and LCD displays by Deshpande and Schwarz (2005). A third commonly used mechanism is the allocation of capacity by the manufacturer to the distributors, as discussed by Cachon and Lariviere (1999a, b). This mechanism is applied to semiconductor manufacturing by Mallik and Harker (2004) and Karabuk and Wu (2005), and to network communications by Niyato and Hossain (2008).

Regarding the distributors' order revision process, Iyer et al. (2003) describe a demand postponement model that arises in the utility, fashion goods, and telecommunications industries, as well as in various service industries. The supplier asks its customers (i.e., the distributors in our problem) to postpone demand in exchange for a discount. As a result, the customers revise their orders based on their available capacity and the amount of the discount. A second example is provided by Chakravarty and

Balakrishnan (2004) in the fashion goods industry. A clothing company—for example, Liz Claiborne—outsources its manufacturing operations to many suppliers, which set capacity aside based on preliminary orders. If final orders from Liz Claiborne's retailers are unexpectedly high, the suppliers do not have enough capacity to meet demand. Thus, the suppliers restrict the amount they deliver, and Liz Claiborne revises its order quantities in real time. A third example of order revision is partial backordering, as applied by Abad (1996) to perishable products. The applications discussed in this paragraph exemplify all the three steps mentioned above.

This paper considers the above capacity allocation and scheduling issues in a make-to-order supply chain. The supply chain includes a manufacturer that produces multiple products in response to orders from several distributors. We study the following three practical and significant *operational coordination issues* that arise in this supply chain when the manufacturer's scheduling capacity is insufficient to meet all the orders it receives.

1. How can a manufacturer effectively incorporate its priorities with respect to profitability into its decisions about capacity allocation among its distributors, while still treating them equitably, and how great are the benefits from doing so?
2. How can the distributors form a stable coalition to share their allocated capacities and divide the resulting profits, and how great are the benefits from doing so?
3. How close to overall coordination of the supply chain does the uncoordinated system come, both when the profit priorities of the manufacturer and distributors are naturally aligned and when they are not?

A further contribution of our work is the consideration of these three interrelated coordination issues jointly, because their overall benefit of coordination is not simply additive. The following three paragraphs review the literature related to these three issues.

There are several studies of capacity allocation issues in single-product problems. Lee et al. (1997) show that allocating capacity in proportion to orders induces overordering. Cachon and Lariviere (1999a) show that allocating capacity based on past sales encourages distributors to increase their sales when demand is low, which benefits manufacturers unfairly. Cachon and Lariviere (1999b) demonstrate how several allocation schemes are vulnerable to false information. Mallik and Harker (2004), Karabuk and Wu (2005), and Durango-Cohen and Yano (2006) study incentive issues in capacity allocation and suggest solutions for poorly aligned incentives. In multiple-product environments, however, basing capacity allocation decisions only on orders received ignores significant effects on scheduling performance. The literature on capacity allocation with multiple products considers capacity allocation issues as manufacturing flexibility (Sethi and Sethi 1990, De Groote 1994, Beach et al. 2000), i.e., the ability to use the same capacity to produce different products.

Manufacturers can allocate capacity either for each individual product, or across products. Using the first approach, Glasserman (1996) describes procedures for choosing base-stock levels and capacity allocations that are asymptotically optimal. However, Bish et al. (2005) argue against allocating capacity within each product because it leads to greater volatility in production levels, higher operational costs, and lower profits. Mechanisms that impose restrictions on specific products within the purchase order provide more control to the manufacturer and less flexibility to the distributors than those that allocate overall capacity across products. Our model defines both allowable order resubmissions for individual products and capacity limits for each distributor across all products. Therefore, our work considers a continuum of relative supply chain power between the manufacturer and the distributors that generalizes capacity allocation practice in many industries.

In revising their orders, distributors may choose to combine their orders with those of other distributors. In our model, this is motivated by the fact that the manufacturer does not know the profitability of each product to each distributor when it allocates capacity and orders to the distributors. As a result, the manufacturer's allocation does not in general maximize the distributors' total profit. Hence, the distributors can potentially benefit from combining their allocated capacity and orders. There are numerous situations where distributors combine their orders in industry (AIS International 2009, All Material Handling 2009, Manhattan Associates 2009). For example, All Material Handling combines orders from up to five distributors into a single order that is shipped as a full container load shipment. Then, the original orders are separated at the arrival port and delivered to their respective final destinations. This consolidated ordering and shipping process allows distributors to save on shipping, ordering, handling, and storage costs. More generally, distributors' consortia exist for the purpose of combining orders in a variety of industries, including office supply, hospitality, furniture, food, restaurant, and janitorial supply (*Cleanlink News* 2007). The advantages of combining orders include access to volume discounts and savings in transportation cost. The literature also includes three streams of research where distributors combine their orders, as now discussed. Besides savings from volume discounts, orders are combined to take advantage of risk pooling in newsvendor centralization games (Hartman et al. 2000, Chen and Zhang 2009a), and orders are combined to reduce total setup costs in economic lot sizing games (van den Heuvel et al. 2007, Chen and Zhang 2009b) and in joint replenishment games (Anily and Haviv 2007, Zhang 2009). Finally, allowing the distributors to combine their orders is consistent with a common assumption in the capacity allocation literature that distributors are not in direct competition with each other (Cachon and Lariviere 1999a, b).

The coordination of operational decisions by the manufacturer and the distributors falls within the research area of *supply chain scheduling* (Hall and Potts 2003). Chen

and Vairaktarakis (2005), Agnetis et al. (2006), Chen and Pundoor (2006), Dawande et al. (2006), and Chen and Hall (2007) study the coordination of scheduling and other operational decisions within supply chains. The coordination issues studied here are similar to those in supply chain scheduling, in that scheduling is coordinated with other important supply chain decisions. Also, the benefits of coordinating the entire supply chain are evaluated exactly, by using optimal algorithms.

This paper is organized as follows. In §2, we present our notation, formally define the problem, and discuss our assumptions. Section 3 models the manufacturer's capacity and order allocation problem as one of scheduling orders to minimize total work-in-process cost on a single production line. In §4, we model the coordination of order quantity decisions among the distributors, subject to their allocated capacity and resubmittable orders, as a cooperative game. We study the existence of fair payoff divisions for coalitions, and we develop algorithms to find them. In §5, assuming that the manufacturer and the distributors coordinate their decisions, we model the entire supply chain and provide a computationally efficient optimal algorithm. Section 6 contains a computational study of the methods developed in the three previous sections. The results provide extensive managerial insights about the value of all three types of coordination discussed above. In §7, we summarize our work and present suggestions for future research. Additional material appears in an electronic companion to this paper that can be found at <http://or.journal.informs.org/>. In the electronic companion, we provide complete proofs of two results in the paper. Extensions to a model where capacity is allocated in integer units are discussed by Liu (2007) and Hall and Liu (2009).

2. Preliminaries

Let $M = \{1, \dots, m\}$ denote the m products produced by the manufacturer, and $N = \{1, \dots, n\}$ denote the n distributors. Let $q = mn$. We follow the common assumption in the capacity allocation literature (Cachon and Lariviere 1999a, b) that only a single order can be processed at any time by the manufacturer. Thus, capacity measures production time. Let $T > 0$ denote the manufacturer's available production capacity. An order from distributor j for one or more units of product i is denoted by ν_{ij} . Let p_{ij} denote the capacity required by order ν_{ij} . We allow the allocation of fractional units of capacity. Let $P_j = \sum_{i \in M} p_{ij}$ and $P_{\max} = \max_j \{P_j\}$.

There are many practical situations where a manufacturer with insufficient capacity fills orders partially (Milner and Kouvelis 2005). There are many other situations where distributors accept only full orders, especially for products with low production cost but high transportation cost (Kasilingam 1999). Therefore, we consider both these alternatives. The manufacturer allocates capacity y_{ij} to order ν_{ij} . If partial orders are allowed, then $0 \leq y_{ij} \leq p_{ij}$; otherwise,

$y_{ij} \in \{0, p_{ij}\}$. Each order consists of a single product, and we assume that the entire order quantity is delivered at the same time, which reduces transportation and stocking costs. Therefore, we consider only schedules where no preemption is allowed in the processing of the orders. However, different orders from the same distributor may be delivered by the manufacturer at different times. Suppose that each order ν_{ij} generates a revenue u_{ij} for the manufacturer and a profit v_{ij} for distributor j . Then, the revenue per unit of capacity from order ν_{ij} is $\bar{u}_{ij} = u_{ij}/p_{ij}$ for the manufacturer, and the profit per unit of capacity is $\bar{v}_{ij} = v_{ij}/p_{ij}$ for distributor j .

An alternative assumption about the ordering and delivery processes is to consider each distributor as issuing only a single *combined order*, which may include multiple products that are delivered together. In this case, the manufacturer defines a weight that measures the scheduling cost associated with each combined order. Thus, the scheduling problem remains the same as the one studied here. However, the model studied in our work is more general, in that it allows *lot splitting* (Trietsch and Baker 1993). In practice, combined orders are often split for delivery purposes to improve customer service, especially for time-sensitive products (Chen 2010).

Given a schedule σ of orders, let $C_{ij}(\sigma)$ denote the completion time of each order ν_{ij} . Let S denote the set of orders scheduled. The manufacturer maximizes its total profit, which is its revenue from the orders processed less its scheduling cost. The scheduling cost is the total weighted completion time, $\sum_{\nu_{ij} \in S} w_{ij} C_{ij}(\sigma)$, where w_{ij} represents either a holding cost per unit time or the value already added to order ν_{ij} . This is a widely used measure of work-in-process inventory cost in manufacturing systems (Pinedo 2002). We assume that the cost of a partially filled order is proportional to its capacity requirement. Thus, for a partially filled order with processing time $y_{ij} < p_{ij}$, the weight of the order is $w_{ij} y_{ij}/p_{ij}$.

Where the schedule is clear from context, we write $C_{ij}(\sigma)$ as C_{ij} . Because $\sum_{\nu_{ij} \in S} w_{ij} C_{ij}(\sigma)$ is a nondecreasing function of order completion time, without loss of generality we consider only schedules without inserted idle time. Smith (1956) shows that the *shortest weighted processing time* (SWPT) rule, where orders are sequenced by nondecreasing ratio of p_j/w_j , minimizes the total weighted completion time in single-machine processing.

We now define the distributors' problem. Each distributor $j \in N$ is allocated capacity $X_j \leq P_j$ by the manufacturer. Let $\bar{X} = \sum_{j \in N} X_j \leq T$. Each order ν_{ij} is given a maximum resubmittable size $\hat{p}_{ij} \leq p_{ij}$ by the manufacturer. Let $\hat{P}_j = \sum_{i \in M} \hat{p}_{ij}$ and $\hat{P}_{\max} = \max_j \{\hat{P}_j\}$. Let $\hat{v}_{ij} = \hat{p}_{ij} v_{ij}/p_{ij}$ be the profit of order ν_{ij} with size $\hat{p}_{ij}$. A distributor may choose its revised orders by itself, subject to its capacity allocation. Alternatively, it may form a coalition with other distributors to share their capacity allocations, codetermine their revised order quantities, and share the resulting profit. For example,

if distributors 1 and 2 form a coalition, then that coalition has total available capacity $X_1 + X_2$.

We model the formation of a coalition among the distributors as a cooperative game. Distributor j chooses a revised order with capacity requirement x_{ij} for product i . After the specification of resubmittable orders by the manufacturer, we have $x_{ij} \leq \hat{p}_{ij}$. If the manufacturer allows partial orders, i.e., $0 \leq x_{ij} \leq \hat{p}_{ij}$, then the distributors' game is a *linear programming game* (Owen 1975). Otherwise, $x_{ij} \in \{0, \hat{p}_{ij}\}$, and each coalition faces a knapsack problem. This distributors' game has apparently not been studied previously, and we term it a *knapsack game*. We assume that a knapsack problem with n items and capacity K can be solved in $O(nK)$ time, using a standard algorithm (Kellerer et al. 2004).

In a cooperative game (Peleg and Sudhölter 2003), a coalition of all the players is the *grand coalition*. Let V denote the maximum profit of the grand coalition. In a particular instance, if there exists a division of V such that no subset of players can improve its total payoff by departing from the coalition, then that division is *fair*. The set of all fair payoff divisions is the *core* of a game. A game with a nonempty core for all instances is *balanced*.

3. Uncoordinated Supply Chain: Manufacturer

In this section, we consider a manufacturer that has received orders that it cannot meet. In §3.1, we study the manufacturer's capacity allocation problem. In §3.2, we study the manufacturer's problem of scheduling the distributors' revised orders.

3.1. Capacity Allocation

We first introduce two simple and widely used rules for capacity allocation, *proportional allocation* (PA) and *linear allocation* (LA), respectively (Laffont 1988, Lee et al. 1997, Cachon and Lariviere 1999b). The PA rule allocates capacity to each distributor proportionately to its capacity requirement:

$$X_j = \min \left\{ P_j, P_j T / \sum_{k \in N} P_k \right\}.$$

The LA rule awards each distributor j its capacity requirement minus the common deduction that satisfies the capacity constraint, or 0 if this difference is negative. If we index the distributors by nonincreasing capacity requirements, i.e., $P_1 \geq \dots \geq P_n$, then

$$X_j = \begin{cases} P_j - \frac{1}{\tilde{n}} \max \left\{ 0, \sum_{k=1}^{\tilde{n}} P_k - T \right\}, & \text{if } j \leq \tilde{n}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\tilde{n}$ is the largest integer no greater than n , such that $P_{\tilde{n}} - \max \{0, \sum_{j=1}^{\tilde{n}} P_j - T\} / \tilde{n} \geq 0$.

These allocation rules are *equitable* between the distributors, in that the capacity allocation is nondecreasing in the order requirement. However, both the PA and LA rules are based solely on order requirements. Because they do not consider revenue or scheduling cost, the resulting capacity allocations could generate small revenues and large costs for the manufacturer. To resolve this problem, we propose the following algorithm, which integrates both scheduling information and equity considerations into the capacity allocation decision.

Proportional Allocation Algorithm (PAAr)

Step 0. Given u_{ij} , p_{ij} , w_{ij} , T , and r .

Step 1. Find a schedule that maximizes the total net profit of the manufacturer, subject to the capacity constraint. Suppose that in this schedule each order v_{ij} is processed at size p'_{ij} .

Step 2. Allocate capacity $X_j = \min \{P_j, P_j T / \sum_{k \in N} P_k\}$ to distributor j , for $j = 1, \dots, n$.

Step 3. Allocate a set of resubmittable orders with total capacity $\hat{P}_j \geq rX_j$ to distributor j , for $j = 1, \dots, n$. Allocate all the orders that appear in the manufacturer's optimal schedule in Step 1, each with a maximum resubmittable size $\hat{p}_{ij} = p'_{ij}$. If the total resubmittable size of those orders is less than rX_j , then allocate other orders in nonincreasing order of revenue per unit of capacity, until $\hat{P}_j \geq rX_j$, where the maximum resubmittable size $\hat{p}_{ij}$ is equal to p_{ij} for each of those allocated orders except for the last one, and is equal to the minimum feasible value satisfying $\hat{P}_j \geq rX_j$ for the last allocated order.

It is important to note that although the capacity of the resubmittable orders can exceed the allocated capacity X_j by a ratio r , the total capacity requirement of the actually resubmitted orders cannot exceed X_j . Hence, all the revised orders can be feasibly scheduled.

The set of resubmittable orders is selected without knowing the distributors' profits. In PAAr, the value of r is a measure of the bargaining power of the distributors, relative to that of the manufacturer. For example, a larger r value provides the distributors more flexibility in revising their orders, hence the manufacturer has less assurance of achieving a high profit level from the resubmitted orders. If r is not large, then the manufacturer, in Step 3 of PAAr, excludes from the set of resubmittable orders those that are less profitable. In practice, either the distributors (Bish et al. 2005) or the manufacturer (Engels et al. 2003) can have dominant bargaining power. Carr and Duenyas (2000) provide an example of the latter case, where a major glass manufacturer allocates capacity on the basis of each individual order. Our definition of r parametrizes the situations in between these two extreme cases.

We now study the manufacturer's scheduling problem with partial orders in Step 1 of PAAr. The first result characterizes the sequence of orders in an optimal schedule for the manufacturer.

LEMMA 1. *For the manufacturer's scheduling problem with partial orders, an optimal schedule processes complete or partial orders in SWPT sequence.*

PROOF. Given a set of complete and partial orders to be processed, the total revenue of those orders is independent of their processing sequence. Thus, the maximum total profit from those orders is found by a schedule that minimizes the total weighted completion time. It follows that, because the weight w_j is reduced proportionately with the capacity allocation of order j , an optimal schedule processes complete or partial orders in SWPT sequence (Smith 1956). $\square$

In view of Lemma 1, we assume for the remainder of this section that the orders are reindexed in SWPT sequence, i.e., $p_1/w_1 \leq \dots \leq p_q/w_q$.

The manufacturer's scheduling problem, which we denote by QP, can be formulated as

$$\max \sum_{j=1}^q \bar{u}_j y_j - \sum_{j=1}^q \left[w_j y_j \left(\sum_{k=1}^j y_k \right) / p_j \right] \quad (1)$$

$$\text{s.t.} \quad \sum_{j=1}^q y_j \leq T, \quad (2)$$

$$0 \leq y_j \leq p_j, \quad j = 1, \dots, q. \quad (3)$$

The objective function (1) maximizes the net profit of the manufacturer. Constraint (2) ensures that the total capacity requirement of the scheduled orders does not exceed the total available capacity. Constraints (3) ensure that each scheduled order is no larger than the corresponding order placed by the distributor. Let L denote the length of a binary encoding of the data for QP.

THEOREM 1. *Problem QP can be solved optimally in $O(m^3 n^3 L)$ time.*

PROOF. The optimality of the SWPT indexing of the variables in problem QP follows from Lemma 1. Observe that the objective function of problem QP is a quadratic function of the form $uy - y^T Qy$, where the matrix Q is positive semidefinite and y^T denotes the transpose of the vector y . It follows that the objective function is concave. The result then follows from Goldfarb and Liu (1993). $\square$

When partial orders are not allowed, we propose the following procedure to solve the manufacturer's scheduling problem in Step 1 of PAAr.

Procedure Man

Input

$p_j, \bar{u}_j, w_j$, for $j = 1, \dots, q$, where $p_1/w_1 \leq \dots \leq p_q/w_q$; T .

Value Function

$F(j, t)$ = maximum profit from orders $1, \dots, j$ achieved by a partial schedule that completes at time t .

Boundary Condition

$$F(0, t) = \begin{cases} 0, & \text{if } t = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Optimal Solution Value

$$\max_{t \in \{0, \dots, T\}} \{F(q, t)\}.$$

Recurrence Relation

$$F(j, t) = \max_{y_j \in \{0, p_j\}} \{\bar{u}_j y_j - w_j t y_j / p_j + F(j-1, t - y_j)\}.$$

In the recurrence relation, y_j has a value of p_j if order j is processed, and a value of 0 if not. In the former case, a profit of $\bar{u}_j y_j$ is earned, a scheduling cost of $w_j t y_j / p_j$ is incurred, and state variables $j-1$ and $t - y_j$ are updated to j and t , respectively.

THEOREM 2. *For the manufacturer's scheduling problem without partial orders, Procedure Man finds a schedule with maximum total profit in $O(mnT)$ time.*

PROOF. The optimality of Procedure Man follows from Lemma 1. In the value function, there are $O(mn)$ possible values of j , and $O(T)$ possible values of t . In the recurrence relation, the maximum profit values resulting from the two possible choices for y_j are compared. Therefore, the overall time complexity of Procedure Man is $O(mnT)$. $\square$

When partial orders are not allowed, the manufacturer's scheduling problem is binary NP-hard, even without the capacity constraint (Engels et al. 2003).

3.2. Scheduling Revised Orders

We now consider the manufacturer's problem of scheduling the distributors' revised orders. Observe from Step 2 of Algorithm PAAr that the total capacity requirement of the revised orders of distributor j must not exceed its allocated capacity X_j . Hence, a feasible schedule is guaranteed and all the revised orders are fully processed. The total revenue is fixed by the manufacturer's requirement to schedule revised orders within the allocated capacity and the set of resubmittable orders. Therefore, maximizing profit is equivalent to minimizing total weighted completion time. We have the following result.

COROLLARY 1. *The manufacturer's problem of scheduling the revised orders to maximize the total profit can be solved in $O(mn \log mn)$ time.*

PROOF. Because all the revised orders are scheduled, revenue is fixed. Therefore, an SWPT sequence minimizes the scheduling cost and maximizes the total profit (Smith 1956). $\square$

4. Uncoordinated Supply Chain: Distributors

In §4.1 we consider as a cooperative game, the distributors' capacity sharing and order revision problem, where the distributors can submit partial orders. In §4.2, we consider the case without partial orders.

4.1. With Partial Orders

We consider the situation where the distributors can resubmit an order that is no larger than the corresponding resubmittable size and may include partial orders with fractional capacity requirements. We first show that the capacity sharing and order revision problem of each distributor or coalition of distributors can be modeled as a linear program (LP). If a coalition contains a set N' of distributors each having allocated capacity X_j , then this problem is as follows:

$$\max \sum_{i \in M} \sum_{j \in N'} \bar{v}_{ij} x_{ij} \quad (4)$$

$$\text{s.t.} \sum_{i \in M} \sum_{j \in N'} x_{ij} \leq \sum_{j \in N'} X_j, \quad (5)$$

$$0 \leq x_{ij} \leq \hat{p}_{ij}, \quad i \in M, j \in N'. \quad (6)$$

The objective function (4) maximizes the total profit of the coalition, where $\bar{v}_{ij}$ is the profit per unit of manufacturing capacity from order v_{ij} for distributor j . Constraint (5) ensures that the total capacity requirement of the revised orders does not exceed the total allocated capacity. Constraints (6) ensure that each revised order is no larger than the corresponding resubmittable size, $\hat{p}_{ij}$. If the manufacturer uses the PA or LA rule, this size is equal to p_{ij} . If the manufacturer implements PAAr, this size is specified in Step 3 for resubmittable orders, and is equal to 0 for unsubmitable orders. The constraints in problem (4)–(6) include only a single constraint (5) with integer coefficients all equal to 1, and simple integer upper bound constraints (6). Therefore, the LP has an optimal integer solution.

The distributors' capacity allocation problem is a *linear programming game* (Owen 1975). In any LP game, some core members (the *Owen set*) can be found by solving the dual problem of the grand coalition's LP. In the distributors' game, there are $mn + 1$ constraints in the LP model of the grand coalition. Letting γ_{ij} , for $i \in M, j \in N$, and γ be the $mn + 1$ dual variables, the dual problem is

$$\min \sum_{i \in M, j \in N} \hat{p}_{ij} \gamma_{ij} + \gamma \sum_{j \in N} X_j \quad (7)$$

$$\text{s.t.} \gamma_{ij} + \gamma \geq \bar{v}_{ij}, \quad i \in M, j \in N, \quad (8)$$

$$\gamma_{ij}, \gamma \geq 0, \quad i \in M, j \in N. \quad (9)$$

The following algorithm finds a core member of the distributors' LP game efficiently.

Algorithm LPCore

Input

$\hat{p}_{ij}$, $\bar{v}_{ij}$, for $i \in M, j \in N$; X_j , for $j \in N$.

Step 1. Find $\bar{v} = \min\{\bar{v}_{ij} : i \in M, j \in N, \sum_{k \in M, l \in N, \bar{v}_{kl} > \bar{v}_{ij}} \hat{p}_{kl} \leq \sum_{j \in N} X_j\}$.

Step 2. Let $\gamma = \bar{v}$, and $\gamma_{ij} = \max\{\bar{v}_{ij} - \bar{v}, 0\}$, for $i \in M, j \in N$. Let the payoff division $(\chi_1, \dots, \chi_n)$ be

$$\chi_j = \sum_{i \in M} \hat{p}_{ij} \gamma_{ij} + X_j \gamma, \quad j \in N. \quad (10)$$

THEOREM 3. Algorithm LPCore finds a core member of the distributors' LP game in $O(mn)$ time.

PROOF. From constraints (8) and (9), the objective function (7) is minimized by setting $\gamma_{ij} = \max\{\bar{v}_{ij} - \gamma, 0\}$. Therefore, in (7), if γ increases by 1, then $\sum_{i \in M, j \in N} \hat{p}_{ij} \gamma_{ij}$ decreases by $\sum_{i \in M, j \in N, \bar{v}_{ij} > \gamma} \hat{p}_{ij}$, and $\gamma \sum_{j \in N} X_j$ increases by $\sum_{j \in N} X_j$. Consequently, from the definition of $\bar{v}$, Step 2 of Algorithm LPCore finds an optimal solution. The core membership of the payoff vector defined by (10) follows from Owen (1975).

Regarding the time complexity of Algorithm LPCore, we show that the value of $\bar{v}$ can be found in Step 1 in $O(mn)$ time. First, from Schönhage et al. (1976), the median of mn elements can be found using $O(mn)$ comparisons. Second, using the definition of $\bar{v}$, it can be checked in $O(mn)$ time whether $v_{ij} \geq \bar{v}$ for any given v_{ij} value. To find $\bar{v}$, we first find the median of all the v_{ij} values, which we denote here by v_M . If $v_M \geq \bar{v}$ (respectively, $v_M < \bar{v}$), then the instance is reduced to consider only the v_{ij} values no greater than (respectively, greater than) v_M and their associated $\hat{p}_{ij}$ values. We continue this procedure until $\bar{v}$ is found. Thus, the overall time complexity of Step 1 is $O(mn + mn/2 + mn/4 + \dots) = O(mn)$. Also, Step 2 requires $O(mn)$ time. Therefore, the overall time complexity of Algorithm LPCore is $O(mn)$. $\square$

For some special classes of LP games, the Owen set covers the entire core (Shapley and Shubik 1972, Kalai and Zemel 1982, Samet and Zemel 1984). However, the following example shows that this is not the case for the distributors' LP game.

EXAMPLE 1. Two distributors 1 and 2 order a single product, with profits 2 and 3 per unit of capacity used, and capacity requirements of their orders 20 and 10, respectively. Previously, the manufacturer has allocated capacity 18 and 9 to distributors 1 and 2, respectively, without restricting the resubmittable orders. Without coordination, distributors 1 and 2 obtain a profit of $2(18) = 36$ and $3(9) = 27$, respectively. If the two distributors form a coalition, then their LP game is

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 \leq 20, \\ & x_2 \leq 10, \\ & x_1 + x_2 \leq 18 + 9 = 27, \end{aligned}$$

where the optimal solution value is 64, with $x_1 = 17$ and $x_2 = 10$. Letting γ_1 , γ_2 , and γ denote the three dual

variables, the dual of this game is

$$\begin{aligned} \min \quad & 20\gamma_1 + 10\gamma_2 + 27\gamma \\ \text{s.t.} \quad & \gamma_1 + \gamma \geq 2, \\ & \gamma_2 + \gamma \geq 3, \\ & \gamma_1, \gamma_2, \gamma \geq 0, \end{aligned}$$

where the unique optimal solution is $\gamma_1 = 0$, $\gamma_2 = 1$, and $\gamma = 2$. Thus, the unique fair payoff division in the Owen set is (36, 28). However, any payoff division $(36 + a, 28 - a)$, where $0 \leq a \leq 1$, is in the core.

Core solutions that are not in the Owen set may be more reasonable than those that are. In Example 1, the payoff division (36, 28) derived from the unique Owen set solution seems to favor distributor 2, which receives all the extra payoff from cooperation. Hence, it is likely that distributor 1 will propose a different payoff division. To evaluate whether this payoff division is in the core, we need to answer the *core membership test*: given an instance of the distributors' game and a payoff vector, is that vector in the core of the instance? In a general LP game, the core membership test is unary co-NP-complete (Fang et al. 2002).

The next two results are interesting for the following reason. In most cooperative games the core membership test is unary co-NP-complete (Deng and Papadimitriou 1994), except for several special cases of the LP game where the Owen set covers the entire core (Shapley and Shubik 1972, Kalai and Zemel 1982, Samet and Zemel 1984). However, in our distributors' game the Owen set is a proper subset of the core, while the core membership test, although binary co-NP-complete, is solvable in pseudopolynomial time.

We first propose a dynamic programming algorithm for the core membership test in the distributors' LP game. For any given payoff vector $(\chi_1, \dots, \chi_n)$ and coalition $N' \subset N$, let the *surplus profit* of N' be defined as the difference between the maximum profit generated by N' and the total payoff paid to N' . We use $F(j, t)$ to denote the maximum surplus profit of a potentially defecting coalition from distributors $\{1, \dots, j\}$, given that those distributors have total allocated capacity t . We assume without loss of generality that the payoff vector $(\chi_1, \dots, \chi_n)$ is efficient, i.e., $\sum_{j \in N} \chi_j$ equals the total profit of the grand coalition; otherwise, the payoff vector is not in the core.

Algorithm LPCoreTest

Input

$\hat{p}_{ij}$, $\bar{v}_{ij}$, for $i \in M$, $j \in N$; X_j , for $j \in N$; $\bar{X}$; and an efficient payoff vector $(\chi_1, \dots, \chi_n)$.

Initialization

For each distributor j , find the maximum profit $\phi(j, t)$ that can be generated using capacity t , for $t = 0, \dots, \min\{\hat{P}_j, \bar{X}\}$, where $\hat{P}_j = \sum_{i \in M} \hat{p}_{ij}$.

Value Function

$F(j, t)$ = maximum surplus profit of any coalition formed

by distributors from $1, \dots, j$, for the given payoff vector $(\chi_1, \dots, \chi_j)$, where distributors $1, \dots, j$ have total allocated capacity t .

Boundary Condition

$F(0, t) = 0$, where $t \geq 0$.

Optimal Solution Value

$F(n, \bar{X})$.

Recurrence Relation

$F(j, t)$

$$= \max \begin{cases} F(j-1, t - X_j), \\ \max_{x_j \in \{0, \dots, \min\{t, \hat{P}_j\}\}} \{\phi(j, x_j) - \chi_j + F(j-1, t - x_j)\}. \end{cases}$$

Core Membership Test

The payoff vector $(\chi_1, \dots, \chi_n)$ is a core member if and only if $F(n, \bar{X}) = 0$.

In the recurrence relation, the first equation excludes distributor j from the coalition being built. The second equation includes distributor j in the coalition and allocates capacity x_j to it, where the value of x_j is chosen to maximize the surplus profit.

THEOREM 4. For a given instance of the distributors' LP game, Algorithm LPCoreTest determines whether a given payoff vector is in the core in $O(n \max\{m, \bar{X}\} \cdot \min\{\bar{X}, \hat{P}_{\max}\})$ time.

PROOF. Observe that the surplus profit of any coalition of the distributors is defined as the difference between the optimal solution of the coalition's optimization problem (4)–(6) and the total payoff to the coalition given by the payoff vector χ . Because this difference is independent of the sequence of the distributors, it is sufficient to consider a single arbitrary sequence. It follows that the surplus profits of all possible state transitions are compared, and hence Algorithm LPCoreTest generates the maximum surplus profit. Therefore, a finding that $F(n, \bar{X}) > 0$ implies the identification of a subset of distributors that will leave the coalition. Alternatively, a finding that $F(n, \bar{X}) = 0$ implies that no such coalition exists, and therefore the payoff vector $(\chi_1, \dots, \chi_n)$ is a core member.

Regarding the time complexity, the initialization step requires $O(mn \min\{\bar{X}, \hat{P}_{\max}\})$ time. In the recurrence relation, there are $O(n)$ possible values of j and $O(\bar{X})$ possible values of t . For each j and t , there are $O(\min\{\bar{X}, \hat{P}_{\max}\})$ values of x_j to enumerate. Therefore, the overall time complexity of Algorithm LPCoreTest is $O(n \max\{m, \bar{X}\} \min\{\bar{X}, \hat{P}_{\max}\})$. $\square$

The following result shows that we cannot expect to find a polynomial time algorithm for the core membership test in the distributors' LP game, unless $P = NP$.

THEOREM 5. The core membership test in the distributors' LP game is binary co-NP-complete, even for a single product.

PROOF. See the electronic companion. $\square$

4.2. Without Partial Orders

We now consider the situation where partial orders are not allowed. This problem is modeled as a knapsack game, defined by Equations (4)–(6) except that Equation (6) is changed to $x_{ij} \in \{0, \hat{p}_{ij}\}$, for $i \in M$, $j \in N'$. Each coalition $N' \subseteq N$ faces a zero-one knapsack problem with knapsack size $\sum_{j \in N'} X_j$, where each item has a size $\hat{p}_{ij}$ and a value $\hat{v}_{ij}$.

We develop a dynamic programming algorithm that finds an integer core member if one exists. The algorithm uses the same definition of surplus profit as Algorithm LPCoreTest.

Algorithm KPCore

Input

$\hat{p}_{ij}$, $\hat{v}_{ij}$, for $i \in M$, $j \in N$; X_j , for $j \in N$; and $\bar{X}$.

Initialization

For each distributor j , find the maximum profit $\phi(j, t)$ that it can achieve using capacity t , for $t \in \{0, \dots, \min\{\hat{P}_j, \bar{X}\}\}$. Find V , by solving the zero-one knapsack problem of the grand coalition.

Value Function

$F(j, t, \bar{X})$ = minimum value of the maximum surplus profit of any coalition formed by distributors $1, \dots, j$, if those distributors have a total allocated capacity t and receive a total payoff $\bar{X}$.

Boundary Condition

$F(1, t, \bar{X})$

$$= \begin{cases} \phi(1, t) - \bar{X}, & \text{if } t < X_1, \text{ or } t \geq X_1 \text{ and } \phi(1, t) \geq \bar{X}, \\ & \text{where } t, \bar{X} \geq 0, \\ 0, & \text{if } t \geq X_1 \text{ and } \phi(1, t) < \bar{X}, \text{ where } t, \bar{X} \geq 0. \end{cases}$$

Optimal Solution Value

$F(n, \bar{X}, V)$.

Recurrence Relation

$F(j, t, \bar{X})$

$$= \min_{x_j \in \{\phi(j, X_j), \dots, \bar{X}\}} \left\{ \max \left\{ \begin{array}{l} F(j-1, t - X_j, \bar{X} - \chi_j) \\ \max_{x_j \in \{0, \dots, \min\{t, \hat{P}_j\}\}} \{ \phi(j, x_j) - \chi_j \\ \quad + F(j-1, t - x_j, \bar{X} - \chi_j) \} \end{array} \right\} \right\}.$$

Core Emptiness Test

If $F(n, \bar{X}, V) > 0$, then the integer core is empty;

If $F(n, \bar{X}, V) = 0$, then the optimal solution value identifies an integer core member.

In the boundary condition, when $t < X_1$, distributor 1's capacity is being used by other distributors. Hence, it must belong to a coalition containing other distributors. Otherwise, the value function takes the larger value achieved by whether distributor j joins a coalition or does not. The recurrence relation first enumerates all possible values of

the payoff χ_j to distributor j . Note that in any fair payoff division, each distributor j receives a payoff no less than its own profit $\phi(j, X_j)$. Then, for each fixed χ_j , the first equation excludes distributor j from the coalition. The second equation includes distributor j in the coalition and allocates capacity x_j to distributor j , where all possible values of x_j are considered.

THEOREM 6. Algorithm KPCore finds an integer core member of an instance of the knapsack game if it has a nonempty integer core, in $O(n\bar{X} \min\{\bar{X}, \hat{P}_{\max}\}V^2)$ time.

PROOF. For fixed χ_j , in order for a coalition to achieve the maximum surplus profit, each distributor in the coalition must use capacity x_j to generate as much profit as possible. Because the surplus profit of the grand coalition is 0, $F(n, \bar{X}, V) \geq 0$. If $F(n, \bar{X}, V) > 0$, then Algorithm KPCore determines that the core of the instance is empty; otherwise $F(n, \bar{X}, V) = 0$, and retracing from the optimal solution value identifies a core member.

In the initialization step, V and the $\phi(j, t)$ values are found in $O(mn\bar{X})$ time. In the recurrence relation, there are $O(n)$ values of j , $O(\bar{X})$ values of t , and $O(V)$ values of $\bar{X}$. To find $F(j, t, \bar{X})$ for fixed j , t , and $\bar{X}$ requires $O(V \min\{\bar{X}, \hat{P}_{\max}\})$ time. Therefore, the recurrence relation requires $O(n\bar{X} \min\{\bar{X}, \hat{P}_{\max}\}V^2)$ computation time, which is also the overall time complexity of Algorithm KPCore. $\square$

The next result shows that the core emptiness test for the knapsack game is intractable. It also shows by example that the knapsack game is not balanced.

THEOREM 7. The problem of deciding whether an instance of the knapsack game has an integer core member is binary NP-hard, even for a single product.

PROOF. See the electronic companion. $\square$

For the knapsack game with integer inputs, the maximum profit generated by each coalition is also integer. However, there might exist a core member with fractional payoffs. Moreover, an instance might have only fractional core members. An example appears in Table 1, where each of the four distributors has a single order. Observe that $(\chi_1, \dots, \chi_4)$ is a fair payoff division. To pay each coalition an integer payoff no less than its own profit, a payoff of at least 7 is required, as with $(\chi'_1, \dots, \chi'_4)$ in the last row of Table 1. However, the grand coalition has a profit of 6. Therefore, no integer core member exists.

Our knapsack game results fundamentally affect the order revision decisions of the distributors in the uncoordinated supply chain. If an instance of the game has a core member, then the grand coalition of the distributors solves its knapsack problem. Otherwise, the grand coalition cannot generate enough payoff to cover every coalition's profit. In this case, it is not possible for subsets of the distributors to form coalitions that are stable (Grieco 1988), as we

Table 1. Instance of the knapsack game with only fractional core members.

j	1	2	3	4
$\hat{v}_j$	3	3	3	3
$\hat{p}_j$	2	2	2	2
X_j	1	1	1	1
χ_j	3/2	3/2	3/2	3/2
χ'_j	2	2	2	1

now show. Suppose that there are several mutually exclusive and exhaustive coalitions of distributors that are stable. This implies that each player in the coalitions receives a sufficient payoff that no players can form a new coalition, including players in other coalitions, that improves their total payoff. However, in this case, combining all the stable coalitions creates a grand coalition that is stable. This contradicts the assumption that there is no core member. It follows that, in an instance with an empty core, each distributor revises its orders individually.

5. Coordinated Supply Chain

In this section, we consider the capacity allocation and scheduling problem when the manufacturer and distributors coordinate their decisions to maximize the profit of the overall supply chain. In the coordinated supply chain, the manufacturer and distributors simultaneously agree on how much should be produced to maximize the profit of the supply chain. This coordination does not require information sharing among the distributors.

THEOREM 8. *An optimal algorithm for problem QP and Procedure Man find a schedule that maximizes the overall supply chain profit, with or without partial orders, respectively.*

PROOF. We replace $\bar{u}_{ij}$ in both QP and Man by $\bar{u}_{ij} + \bar{v}_{ij}$. The remainder of the proof follows that of Theorems 1 and 2. $\square$

We consider a *supply chain game* consisting of one manufacturer and n distributors, where each distributor j has allocated capacity X_j and a set of resubmittable orders. Note that there is an instance of the distributors' game corresponding to each instance of the supply chain game.

For a hypothetical coalition consisting of the manufacturer and some but not all of the distributors, the total profit of the coalition depends on the revised order decisions of distributors that are not in the coalition. This is because both sets of orders are processed using the same resources. Therefore, among possible coalitions containing the manufacturer, we consider only the manufacturer by itself and the grand coalition. In the supply chain game, each coalition and its maximum profit are defined as follows. Let R denote the maximum profit of the grand coalition of the supply chain game, i.e., the maximum profit

achieved by the coordinated supply chain. Let U_M denote the profit of the manufacturer outside of any coalition. Consequently, if the corresponding instance of the distributors' game has a nonempty (respectively, an empty) core, then U_M is the profit achieved by the manufacturer, assuming that the distributors jointly (respectively, individually) revise their orders. The reason for defining U_M in this way is that under supply chain coordination, the manufacturer can access the distributors' information and predict each distributor's behavior if it leaves the grand coalition. Because all other coalitions contain only distributors, their maximum profits are the same as in the corresponding instance of the distributors' game.

We now describe our balancedness, core emptiness test, and core membership test results for the supply chain game.

COROLLARY 2. *For the supply chain game with partial orders:*

- (a) *the game is balanced;*
- (b) *Algorithm LPCore finds a core member of the game in $O(mn)$ time;*
- (c) *Algorithm LPCoreTest determines whether a given payoff vector is in the core of a given instance in $O(n \max\{m, \bar{X}\} \min\{\bar{X}, P_{\max}\})$ time; and*
- (d) *the core membership test of the supply chain game is binary co-NP-complete, even for a single product.*

PROOF. (a) The maximum profit R of the grand coalition is no less than that of the uncoordinated supply chain where the distributors coordinate. Because the corresponding distributors' LP game is balanced, (a) holds.

(b) To find a core member, we first apply Algorithm LPCore to find a core member of the corresponding distributors' LP game, and then pay the manufacturer the remaining payoff, U_M , from R . Thus, the payoff division is fair, and hence (b) follows from Theorem 3.

(c) For the core membership test, compared with the corresponding distributors' LP game, the only additional work requires testing whether the payoff to the manufacturer is no less than U_M . Thus, (c) follows from Theorem 4.

(d) For each order, we can define both the revenue and the work-in-process cost for the manufacturer as 0. This makes the supply chain game equivalent to the distributors' LP game, hence (d) follows from Theorem 5. $\square$

COROLLARY 3. *For the supply chain game without partial orders:*

- (a) *the game is unbalanced;*
- (b) *Algorithm KPCore finds an integer core member, if one exists, in $O(n\bar{X} \min\{\bar{X}, P_{\max}\}(R - U_M)^2)$ time; and*
- (c) *it is binary NP-hard to decide whether an instance of the knapsack game has an integer core member, even for a single product.*

PROOF. (a) If we define the revenue of each order and the work-in-process cost for the manufacturer as 0, then the supply chain game is equivalent to the distributors'

knapsack game. Hence, the proof follows from the example instance in the proof of Theorem 7 in the electronic companion.

(b) For the core emptiness test, we apply Algorithm KPCore. From Theorem 6, if $F(n, \bar{X}, R - U_M) > 0$, then the integer core is empty; otherwise, retracing from the optimal solution value identifies a core member.

(c) The result follows from Theorem 7. $\square$

It is straightforward to observe that the supply chain game and the distributors' game have the following relationship.

COROLLARY 4. *For a given instance of the supply chain game, if a payoff vector is in the core, then the part of the payoff vector that applies to the distributors is also in the core of the corresponding instance of the distributors' game. If an instance of the distributors' game has a nonempty core, then the corresponding instance of the supply chain game also has a nonempty core.*

6. Computational Study

In this section, we use computational testing on randomly generated data to evaluate the benefit that arises from the three types of coordination discussed above. Following the guidelines established by Hall and Posner (2001), (a) we generate a wide range of parameter specifications for robustness, (b) all the parameters can be rescaled together without significantly affecting performance, and (c) the experimental design varies only the parameters that may affect the analysis.

We choose the number of distributors as $n = 20$, and the number of products as $m \in \{5, 10, 20, 50, 100\}$. The number of products directly affects the distributors' flexibility in revising their orders, especially when the distributors do not coordinate or partial orders are not allowed. The number of distributors also affects their flexibility in revising their orders when they coordinate. However, because the grand coalition of distributors typically has a large number of orders, the distributors already have a lot of flexibility and therefore the impact of the value of n is not significant, as a preliminary computational study verified. That study also suggested that the range of capacity requirements (p_{ij}), and the availability of manufacturer's capacity ($T / \sum_{ij} p_{ij}$), do not significantly affect the value of coordination. Therefore, we do not vary these parameters in our main computational study.

Capacity requirements are generated using $p_{ij} \sim U[1, \dots, 50]$. The manufacturer's total capacity is $T = 0.75 \sum_{ij} p_{ij}$. With this value of T , from the definition of PAAR, the maximum meaningful value of r in PAAR is $1/0.75 \approx 1.33$. Therefore, we consider $r \in \{r_1, r_2, r_3\}$, where $r_1 = 1.00$, $r_2 = 1.15$, and $r_3 = 1.33$. The revenue u_{ij} of order v_{ij} is generated as $u_{ij} = \bar{u}_{ij} p_{ij}$, where $\bar{u}_{ij} \sim U[1, \dots, 10]$. The profit v_{ij} of order v_{ij} is generated using $v_{ij} = \bar{v}_{ij} p_{ij}$, as follows.

1. When v_{ij} and u_{ij} are independent, we generate $\bar{v}_{ij} \sim U[1, \dots, 10]$.

2. When v_{ij} and u_{ij} are positively related, we generate

$$\bar{v}_{ij} \sim \begin{cases} U[1, 3], & \text{if } \bar{u}_{ij} = 1, \\ U[\bar{u}_{ij} - 1, \bar{u}_{ij} + 1], & \text{if } 2 \leq \bar{u}_{ij} \leq 9, \\ U[8, 10], & \text{if } \bar{u}_{ij} = 10, \end{cases}$$

which gives a correlation coefficient of 0.946.

3. When v_{ij} and u_{ij} are negatively related, we generate

$$\bar{v}_{ij} \sim \begin{cases} U[8, 10], & \text{if } \bar{u}_{ij} = 1, \\ 11 - U[\bar{u}_{ij} - 1, \bar{u}_{ij} + 1], & \text{if } 2 \leq \bar{u}_{ij} \leq 9, \\ U[1, 3], & \text{if } \bar{u}_{ij} = 10, \end{cases}$$

which gives a correlation coefficient of -0.946 .

These three types of correlation between per order revenues for the manufacturer and per order profits for the distributors model practical situations where the manufacturers and the distributors have neutrality, consistency, or conflict of interest, respectively. The type of correlation significantly affects the value of the three types of supply chain coordination.

To avoid overemphasizing the role of scheduling costs, we assume that no order generates a negative payoff for the manufacturer even if it is completed at the latest possible time, T . This is implemented by generating $w_{ij} = \bar{w}_{ij}/T$, where $\bar{w}_{ij} \sim U[1, \dots, u_{ij}]$.

For each of the 15 possible combinations of the parameters described above, we randomly generate 10 problem instances. For each of those 150 instance, five instances of the knapsack game are generated under the five capacity allocation rules implemented by the manufacturer. We find that 747 out of these 750 knapsack game instances have a nonempty core. For each of the three instances with empty integer core, we identify payoff divisions under which the maximum surplus profit of coalitions is only 1. Therefore, we consider distributors' and supply chain coordination for all the randomly generated instances.

The parameters that have a potentially significant effect on the value of coordination are listed in Table 2. Sections 6.1, 6.2, and 6.3 discuss the effects of these variables on the value of manufacturer's coordination, distributors' coordination, and overall supply chain coordination, respectively. We use analysis of variance (ANOVA) to test significance at a level of $\alpha = 0.01$. In the ANOVA Tables, 3, 5, and 7, we report several statistics. Here, R^2 is the proportion of variation in the dependent variable that is explained by the independent variables; MS represents the between-group variation; the F value is the test statistic computed from computational results; the F critical value is the value that the F value must exceed in order to reject

Table 2. Independent variables to evaluate coordination.

Factors	No. of levels	Levels
m (number of products)	5	5, 10, 20, 50, 100
UV (correlation of orders' revenues for the manufacturer and profits for the distributors)	3	1: independent 2: positively correlated 3: negatively correlated
I (whether partial orders are allowed)	2	1: no 2: yes
A (allocation rules)	5	1: PA 2: LA 3: PAAr ₁ , $r_1 = 1.00$ 4: PAAr ₂ , $r_2 = 1.15$ 5: PAAr ₃ , $r_3 = 1.33$
C (whether distributors coordinate)	2	1: no 2: yes

the null hypothesis that an independent variable does not significantly affect the dependent variable; and the p value is the probability that the F value will be at least as extreme as the current value under the assumption that the independent variable has no impact on the dependent variable.

6.1. Manufacturer's Coordination

To evaluate the benefit of coordination by the manufacturer, we consider five independent variables: A , m , UV , I , and C , and four interactions: $A * m$, $A * UV$, $A * I$, and $A * C$. The dependent variable is the ratio of the manufacturer's profit under each allocation rule to the manufacturer's maximum profit, i.e., the manufacturer's profit from the optimal schedule found in Step 1 of PAAr. The ANOVA and sensitivity analysis results are summarized in Tables 3 and 4, respectively. On average, the PA and LA rules perform similarly, and PAAr generates 10.66% more profit for the manufacturer than the mean result from the PA and LA rules. We have four main findings for the performance of the allocation rules studied.

First, the relative improvement offered by PAAr for small r values increases as the number of products

Table 3. ANOVA results for manufacturer's coordination: $R^2 = 0.9332$.

Source	MS	F value	F critical	p value
A	2.1425	2187.52	3.33	<0.0001
m	0.0358	36.56	3.33	<0.0001
UV	12.1086	12,363.10	4.61	<0.0001
I	0.7139	728.89	6.64	<0.0001
C	0.1032	105.37	6.64	<0.0001
$A * m$	0.0364	37.21	2.01	<0.0001
$A * UV$	0.6966	711.21	2.52	<0.0001
$A * I$	0.1138	116.21	3.33	<0.0001
$A * C$	0.0240	24.48	3.33	<0.0001

Table 4. Sensitivity analysis for manufacturer's coordination.

Source	$A = PA$	$A = LA$	$A = PAAr_1$	$A = PAAr_2$	$A = PAAr_3$
$m = 5$	0.8181	0.8162	0.9067	0.8809	0.8707
$m = 10$	0.8171	0.8168	0.9406	0.9056	0.8632
$m = 20$	0.8189	0.8187	0.9619	0.9206	0.8509
$m = 50$	0.8194	0.8194	0.9762	0.9265	0.8393
$m = 100$	0.8207	0.8208	0.9805	0.9271	0.8379
$UV = 1$	0.8251	0.8248	0.9553	0.9171	0.8612
$UV = 2$	0.9756	0.9750	0.9790	0.9805	0.9803
$UV = 3$	0.6557	0.6554	0.9252	0.8389	0.7157
$I = 1$	0.8164	0.8157	0.9409	0.8814	0.8235
$I = 2$	0.8213	0.8211	0.9655	0.9429	0.8813
$C = 1$	0.8181	0.8172	0.9368	0.9059	0.8478
$C = 2$	0.8196	0.8196	0.9696	0.9185	0.8571

increases. However, for large r values, the relative improvement offered by PAAr decreases as the number of products increases. The algorithm benefits the manufacturer by implicitly forcing the distributors to revise their orders in the manufacturer's favor. For strong manufacturer's bargaining power represented by small r values, this effect is greater with more orders. However, for large r values, this effect becomes weaker when more orders are available for the distributors to revise within their allocated capacity limits.

Second, PAAr provides more benefit when $UV = 3$; for example, when $r = 1.15$, 27.97% additional profit, compared with 11.17% and 0.53% additional profit when $UV = 1$ and 2, respectively. When the correlation between order revenues and distributors' profits is negative, it is more important for the manufacturer that the distributors revise their orders in a way that favors the manufacturer, which PAAr encourages.

Third, PAAr provides more benefit when partial orders are allowed; for example, when $r = 1.15$, 14.82% more profit compared with 8.01% additional profit without partial orders. With partial orders, the distributors have more flexibility to revise their orders and improve their profits. Thus, the manufacturer's order specification in Step 3 of PAAr is more important.

Fourth, PAAr provides more benefit when the distributors coordinate; for example, when $r = 1.15$, 12.07% more profit compared with 10.80% additional profit without the distributors' coordination. The reason is that coordination by the distributors enables them to select more orders, which benefits the entire supply chain.

However, PAAr does not outperform the PA and LA rules for all the instances tested. Compared with the PA rule, PAAr provides less profit for the manufacturer in 9.83% of instances when $r = 1.00$, in 6.33% of instances when $r = 1.15$, and in 4.83% of instances when $r = 1.33$. The majority of instances for which the PA rule outperforms PAAr have small numbers of products and $UV = 2$. When the correlation between order revenues and distributors' profits is positive, PAAr helps the manufacturer less.

Table 5. ANOVA results for distributors' coordination: $R^2 = 0.7710$.

Source	MS	F value	F critical	p value
A	0.0048	25.42	3.33	<0.0001
m	0.2265	1,190.73	3.33	<0.0001
UV	0.0014	7.25	4.62	<0.0001
I	0.0250	131.65	6.65	<0.0001

In Step 3 of PAAr, some resubmittable orders are selected in nonincreasing order of revenue per unit of capacity. This results in the selection of resubmittable orders with small sizes and large weights, creating a high scheduling cost for the manufacturer. Also, when r is small, PAAr specifies a total maximum resubmittable size for each distributor that is close to its allocated capacity. When m is small and partial orders are not allowed, it might be difficult for a distributor to select orders that utilize most of its available capacity. This also reduces the manufacturer's profit.

6.2. Distributors' Coordination

To evaluate the benefit of coordination by the distributors under various parameter settings, we study the four independent variables A , m , UV , and I . The dependent variable is the ratio of the distributors' total profit with coordination over that without coordination. The ANOVA and sensitivity analysis results are shown in Tables 5 and 6, respectively. On average, coordination generates 2.58% more profit for the distributors. We have four main findings.

First, coordination between the distributors is more valuable when the manufacturer uses PAAr, compared with using the PA and LA rules, especially for small r values. By implicitly forcing the distributors to follow the manufacturer's schedule, PAAr potentially hurts the distributors' profit, especially when the manufacturer's and distributors' priorities conflict. However, coordination by the distributors mitigates this effect.

Table 6. Sensitivity analysis for distributors' coordination.

Source	Ratio
A = PA	1.0227
A = LA	1.0237
A = PAAr ₁	1.0320
A = PAAr ₂	1.0277
A = PAAr ₃	1.0228
m = 5	1.0708
m = 10	1.0322
m = 20	1.0152
m = 50	1.0066
m = 100	1.0042
UV = 1	1.0258
UV = 2	1.0241
UV = 3	1.0274
I = 1	1.0299
I = 2	1.0217

Second, coordination generates more profit for the distributors when the number of products is relatively small; for example, 0.42% additional profit for 100 products compared to 7.08% more profit for 5 products. As m decreases, each distributor has fewer orders to revise and thus utilizes its allocated capacity less effectively. Consequently, the sharing of capacity allocation through coordination by the distributors becomes more valuable.

Third, more profit is achieved by the distributors' coordination when $UV = 3$, specifically 2.74% additional profit compared with 2.58% and 2.41% additional profit when $UV = 1$ and 2, respectively. This means that failure to coordinate hurts the distributors more when the priorities of the manufacturer and the distributors conflict.

Fourth, coordination is more valuable when partial orders are not allowed, specifically 2.99% more profit compared with 2.17% additional profit with partial orders. When partial orders are not allowed, each distributor faces a 0–1 knapsack problem with a small number of items, hence the allocated capacity may be used ineffectively.

If the distributors know the overall capacity T , then they can coordinate their orders before first submitting them. If the manufacturer uses either the PA rule or the LA rule, then there is no advantage for the distributors from doing this, because all the orders are considered. On the other hand, if the manufacturer uses the PAAr allocation rule, then coordination among the distributors before the first order submission can generate additional profit for the distributors. How this profit should be divided among the distributors is not clear, however, because the capacity allocated to each coalition is not defined. Therefore, a capacity allocation rule such as PA needs to be used to determine a fair payoff division. Our computational results show that coordination by the distributors before the first order submission generates 16.71%, 8.00%, and 0.32% more profit on average for them, compared to coordination after capacity allocation using the PAAr₁, PAAr₂, and PAAr₃ rules, respectively.

6.3. Supply Chain Coordination

The independent variables studied are A , m , UV , I , and C . The dependent variable is the ratio of the total profit of the uncoordinated supply chain over that of the coordinated supply chain. The ANOVA and sensitivity analysis results are summarized in Tables 7 and 8, respectively. On average, supply chain coordination generates 3.44% more profit. We have four main findings.

First, supply chain coordination is more valuable when, in the uncoordinated supply chain, the manufacturer uses PAAr with small values of r to allocate capacity. For small r values, the distributors achieve limited profit, and thus the uncoordinated supply chain is less efficient. However, when $r = 1.33$, PAAr achieves the largest profit for the uncoordinated supply chain, which is 97.29% of that of the coordinated supply chain.

Table 7. ANOVA results for supply chain coordination:
 $R^2 = 0.6701$.

Source	MS	F value	F critical	p value
A	0.0352	183.94	3.32	<0.0001
m	0.0623	325.01	3.32	<0.0001
UV	0.2518	1,314.31	4.61	<0.0001
I	0.0002	1.07	6.64	0.0215
C	0.2689	1,403.74	6.64	<0.0001

Second, supply chain coordination is relatively more valuable for instances with few products; for example, providing on average 2.59% more profit for 100 products and 5.33% more profit for 5 products. As m decreases, the distributors have fewer orders to revise within their allocated capacity, and therefore they utilize their allocated capacity less effectively.

Third, supply chain coordination is less valuable when the order revenues for the manufacturer and the profits for the distributors are positively correlated (i.e., $UV = 2$), providing on average 1.52% additional profit compared with 4.65% and 4.17% more profit when $UV = 1$ and 3, respectively. Because the manufacturer and the distributors naturally make consistent choices, even uncoordinated decisions are effective, which reduces the benefit of coordination. This means that, when the priorities of the manufacturer and distributors are naturally aligned, the supply chain is almost perfectly coordinated by the independent actions of the manufacturer (on the one hand) and the distributors (on the other hand). This can be viewed as a positive outcome because effective supply chain coordination does not require the manufacturer to cooperate with the distributors. However, if $UV \in \{1, 3\}$, then cooperation between the manufacturer and the distributors is more important.

Fourth, supply chain coordination is more valuable when, in the uncoordinated supply chain, the distributors do not coordinate either, providing on average 4.49% more profit

compared with 2.40% more profit when the distributors coordinate. Without coordination by the distributors, there are $n + 1$ suboptimal decisions, but this number is reduced to 2 if the distributors coordinate. Also, from the results in §§6.1 and 6.2, coordination by the distributors benefits both the manufacturer and the distributors.

7. Conclusions

This paper is among the first that simultaneously consider three significant operational coordination issues that arise in two-stage supply chains with a manufacturer and several distributors. First, we evaluate the cost savings from integrating scheduling information into capacity allocation decisions. Second, we evaluate the additional profit the distributors can earn by sharing their capacity allocations and coordinating their revised orders. Third, we evaluate the additional profit that becomes available to the overall supply chain when the manufacturer and the distributors coordinate their decisions. We also provide insights, based on sensitivity analysis, about when coordinating decisions is likely to be more valuable.

Overall, this paper provides exact evaluations of the trade-offs involved in the three coordination issues. The three types of coordination are all valuable, and the benefit is typically between 1% and 15%. In general, the three types of coordination are especially valuable when the manufacturer and the distributors have conflicting priorities between orders. Coordination is also more valuable when the manufacturer's profit margins are small, in which case scheduling cost savings become relatively more significant.

A natural extension is to consider setup times that occur between the processing of different products. Such setup times arise in various industries (Pinedo 2002), but many scheduling problems with setup times are difficult to solve optimally (Potts and Kovalyov 2000). We conducted a computational study to evaluate the benefit of coordination in the presence of setup times. Our computational results indicate that the benefits from the three types of coordination are on average slightly greater in the presence of setup times than those in the computational study described in §6.

There are several interesting directions for further research. First, the problem of testing whether instances of the knapsack game have fractional core members remains open. Second, because several of our algorithms require pseudopolynomial time, it will be useful to develop effective heuristics to solve instances with large data values. Third, other measures of scheduling cost besides the total weighted completion time can be considered. Also, other operational decisions such as distribution can be incorporated into the supply chain considered. Furthermore, the supply chain studied in this paper, which consists of one manufacturer and several distributors, can be generalized to consider multiple competing manufacturers. Finally, the impact of strategic issues, such as deliberate overordering, on the manufacturer's and distributors' coordination

Table 8. Sensitivity analysis for supply chain coordination.

Source	Ratio
A = PA	0.9718
A = LA	0.9713
A = PAA r_1	0.9544
A = PAA r_2	0.9661
A = PAA r_3	0.9729
m = 5	0.9506
m = 10	0.9649
m = 20	0.9713
m = 50	0.9746
m = 100	0.9751
UV = 1	0.9561
UV = 2	0.9855
UV = 3	0.9604
C = 1	0.9578
C = 2	0.9767

decisions, can be considered. In conclusion, we hope that our work will encourage further research that applies ideas from cooperative game theory to improve operational decision making in supply chains.

8. Electronic Companion

An electronic companion to this paper is available as part of the online version that can be found at <http://or.journal.informs.org/>.

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